US011796128B1

(12) **United States Patent**
Ma et al.

(10) **Patent No.: US 11,796,128 B1**
(45) **Date of Patent: Oct. 24, 2023**

(54) **TWO-IN-ONE MULTI-FUNCTIONAL MAGNETIC BRACKET**

(71) Applicants: **Wenhong Ma**, Guangdong (CN); **Jiawei Gao**, Guangdong (CN)

(72) Inventors: **Wenhong Ma**, Guangdong (CN); **Jiawei Gao**, Guangdong (CN)

(73) Assignee: **Guangdong Shuowei Technology Co., Ltd.**, Dongguan (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/213,905**

(22) Filed: **Jun. 26, 2023**

(51) **Int. Cl.**
F16M 13/02 (2006.01)
H04M 1/04 (2006.01)
H01F 7/02 (2006.01)

(52) **U.S. Cl.**
CPC ***F16M 13/022*** (2013.01); ***H01F 7/0205*** (2013.01); ***H04M 1/04*** (2013.01)

(58) **Field of Classification Search**
CPC F16M 13/022; H01F 7/0205; H04M 1/04; F16B 47/00; F16B 2/08; A45F 5/02
USPC 248/683, 206.5, 309.4, 292.12; 455/575.1
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,164,126	B1 *	10/2015	Rodda	G01R 1/04
9,548,782	B1 *	1/2017	Jeong	B60R 11/02
9,723,910	B2 *	8/2017	Due	A45F 5/10
10,419,054	B1 *	9/2019	VanTassell	H04M 1/04
10,760,732	B1 *	9/2020	Koh	F16M 13/02
11,265,035	B1 *	3/2022	Zhang	H04B 1/3877
11,545,819	B2 *	1/2023	Fan	H02J 7/02
11,658,694	B1 *	5/2023	Wang	H04B 1/3888 455/575.8
2015/0382489	A1 *	12/2015	Sorensen	F16M 11/40 24/3.1

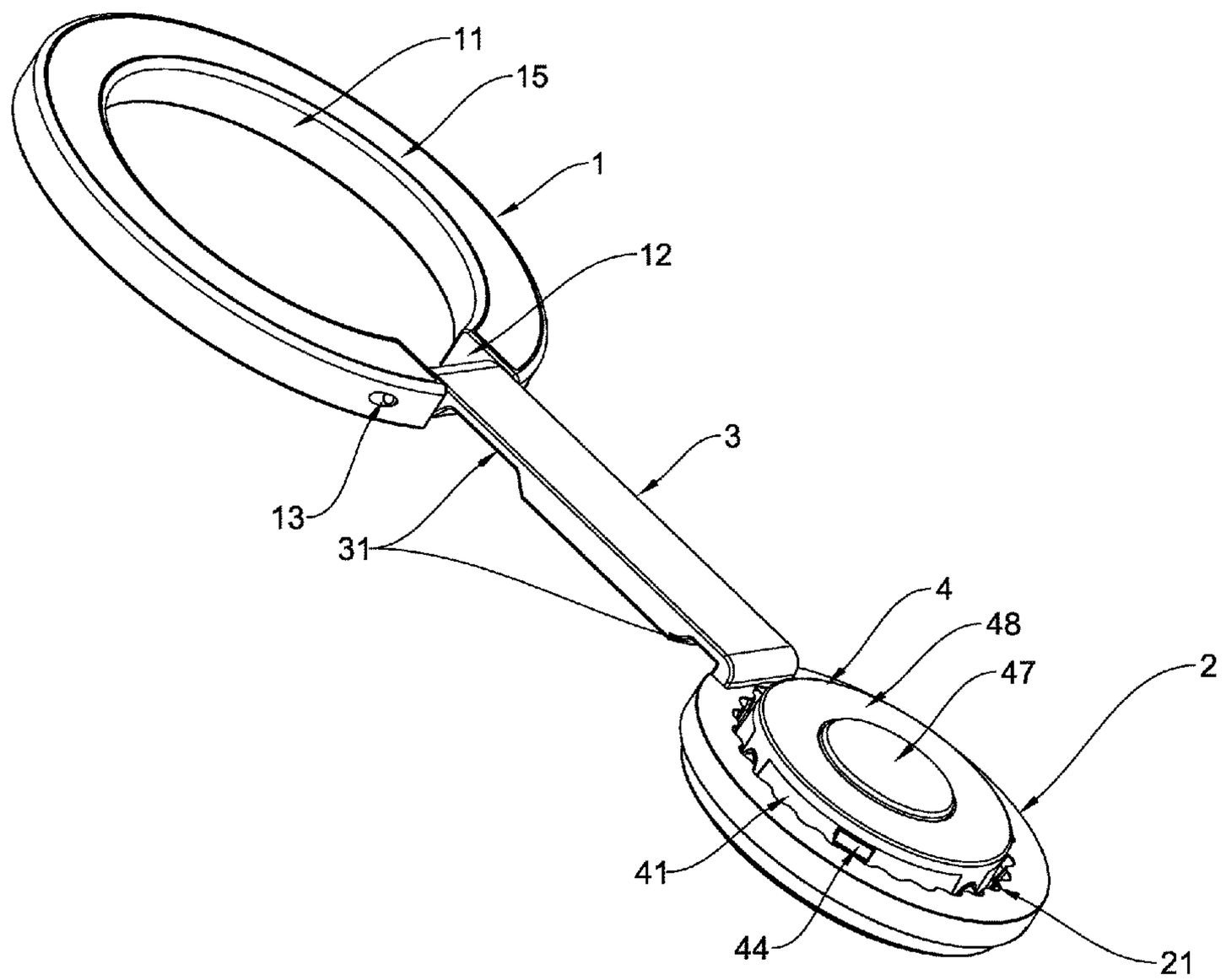
* cited by examiner

Primary Examiner — Todd M Epps

(57) **ABSTRACT**

A multi-functional magnetic bracket includes a first bracket with a magnet disposed therein to magnetically attach the first bracket to mobile phone; a second bracket movably connected to the first bracket through a connecting arm, the second bracket and a hinging arm being accommodated to inside of the first bracket, and a ring and a tooth slot opened in the second bracket; and a connector disposed in the second bracket and combined with the second bracket to form a multi-platform bracket for mounting the second bracket on different scenarios and platforms. The connector includes a base of which a surface is adhered to the platform, a limiting buckle disposed in the base and of which one end extends to a side of the second bracket, and a spring bead disposed in the base and of which one end extends to inside of the tooth slot.

10 Claims, 7 Drawing Sheets



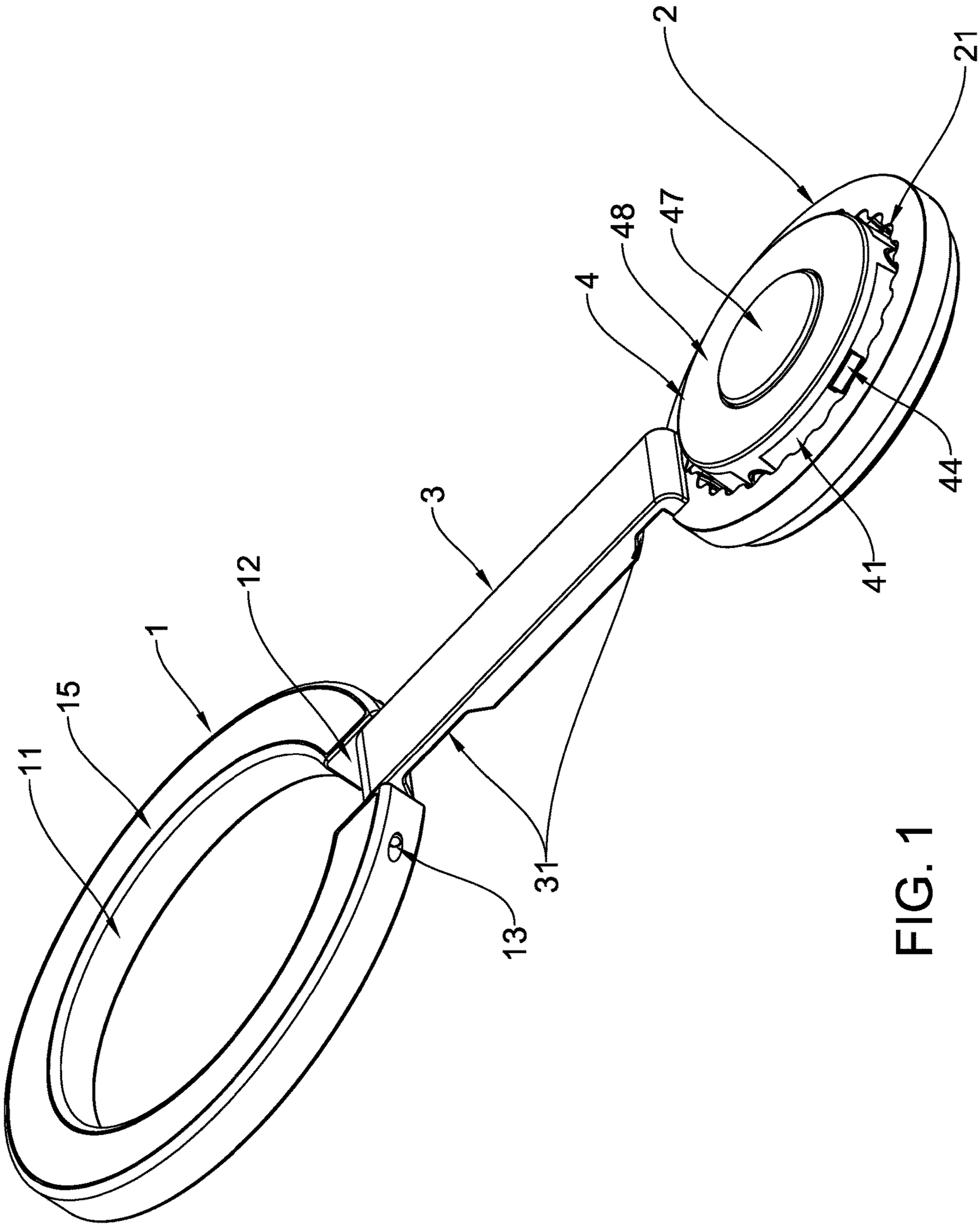


FIG. 1

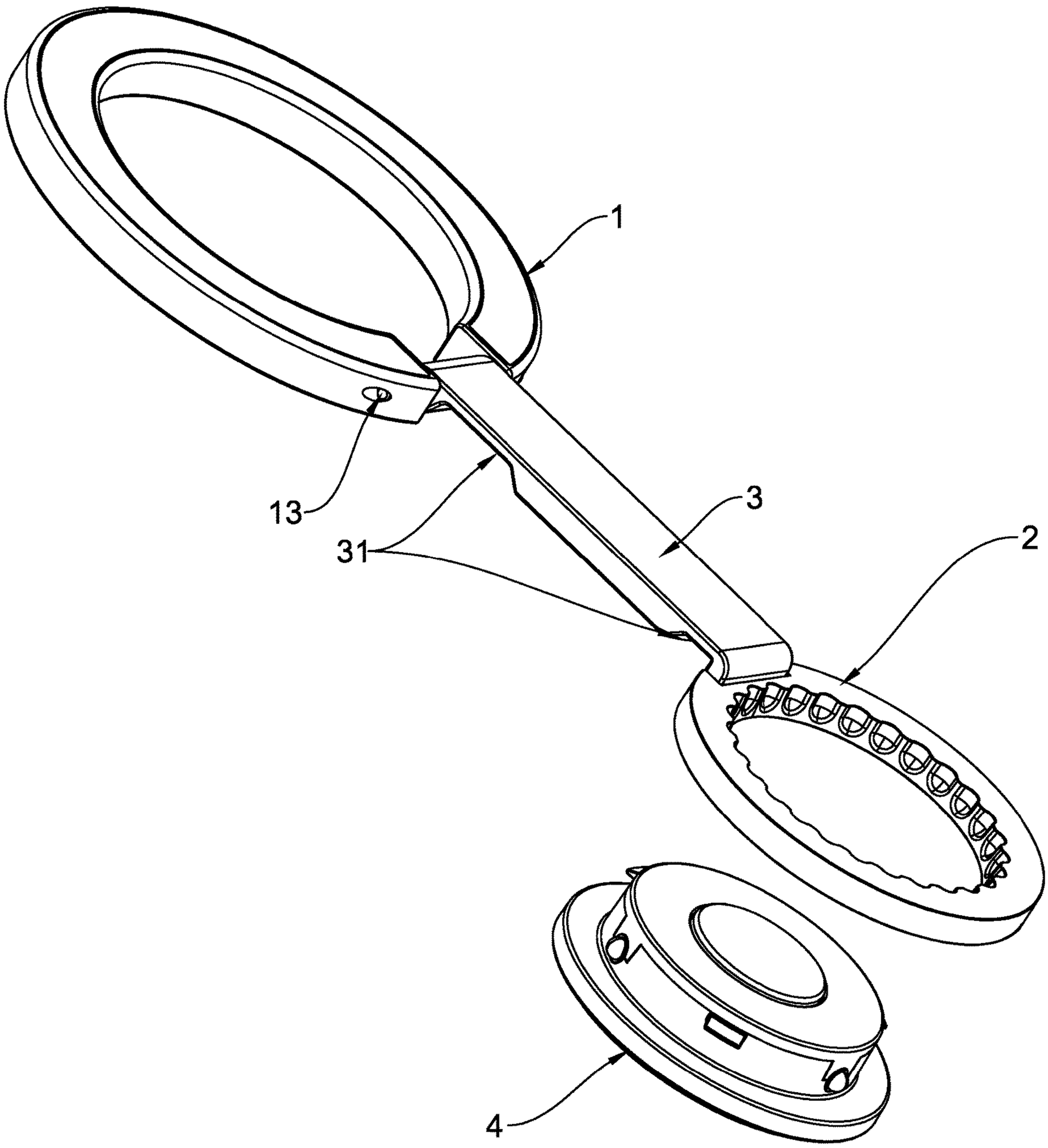


FIG. 2

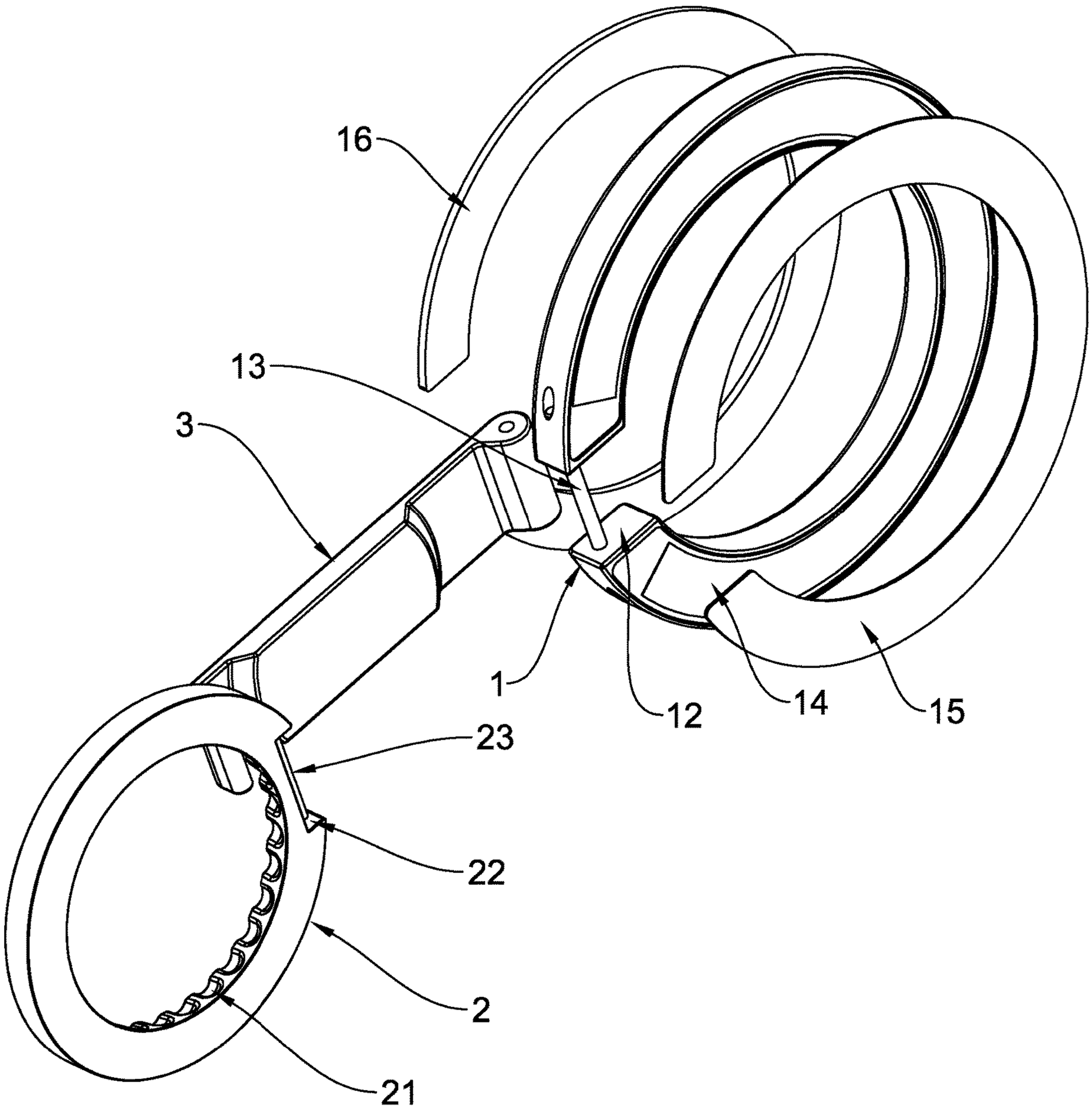


FIG. 3

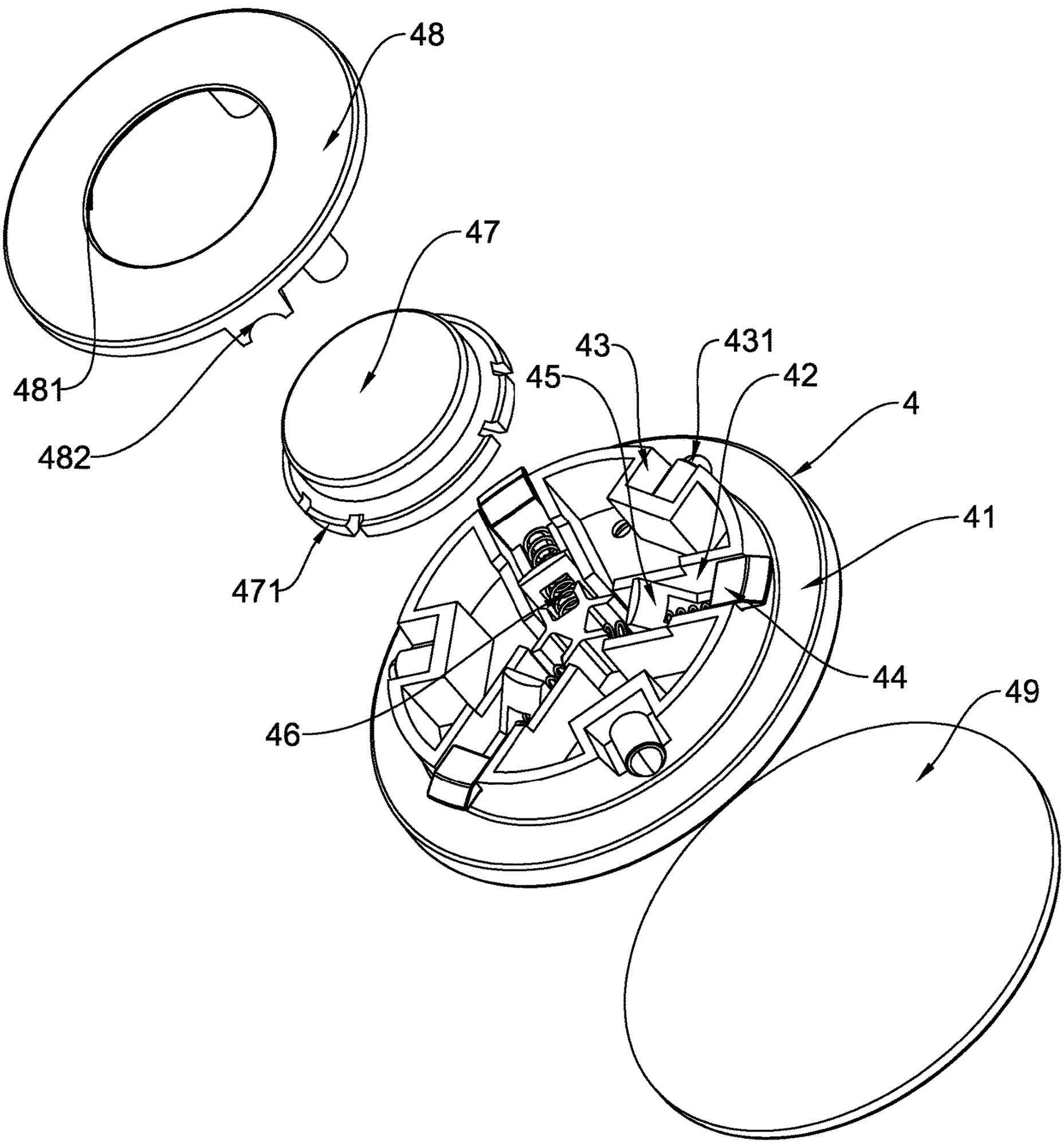


FIG. 4

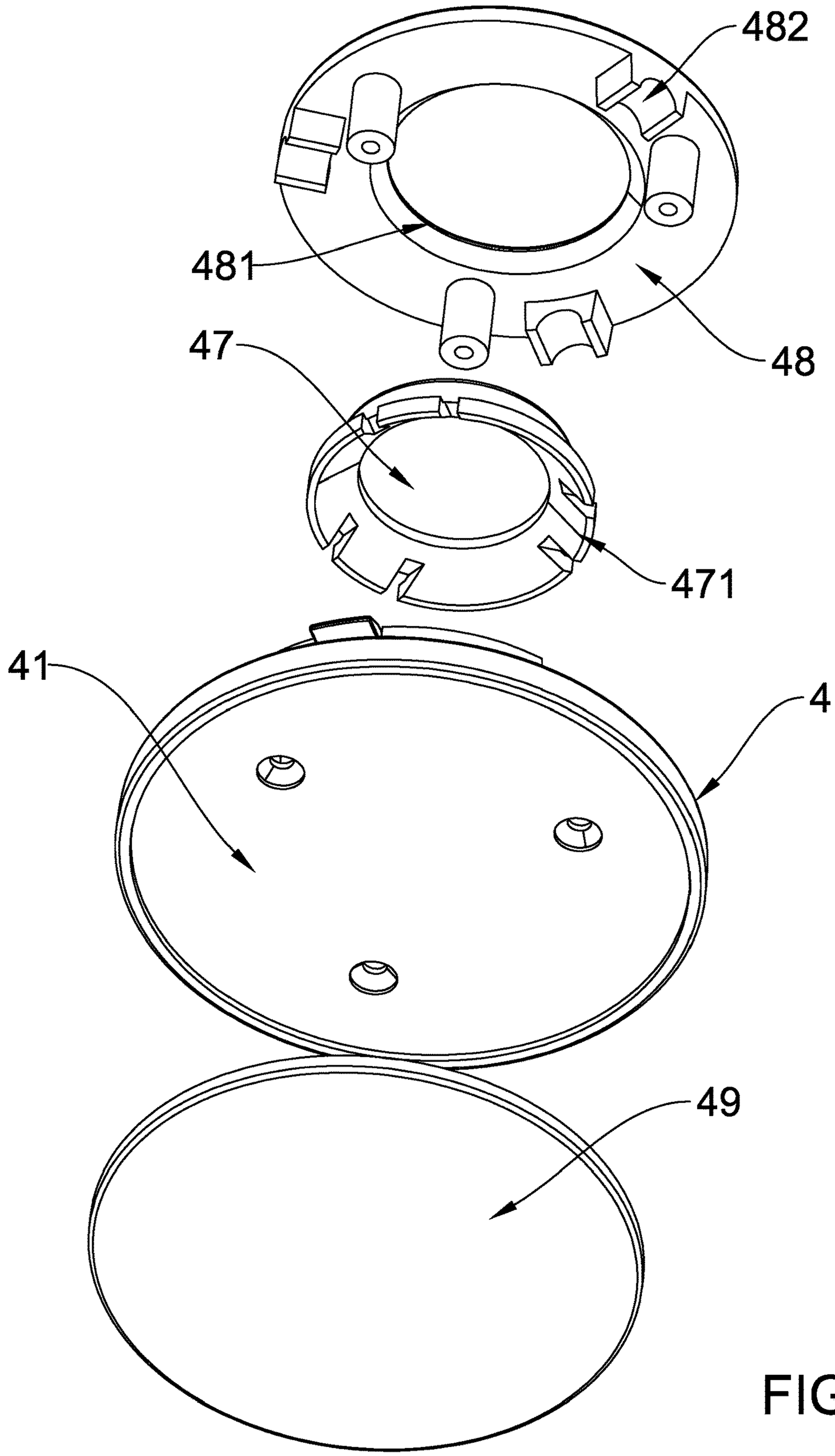


FIG. 5

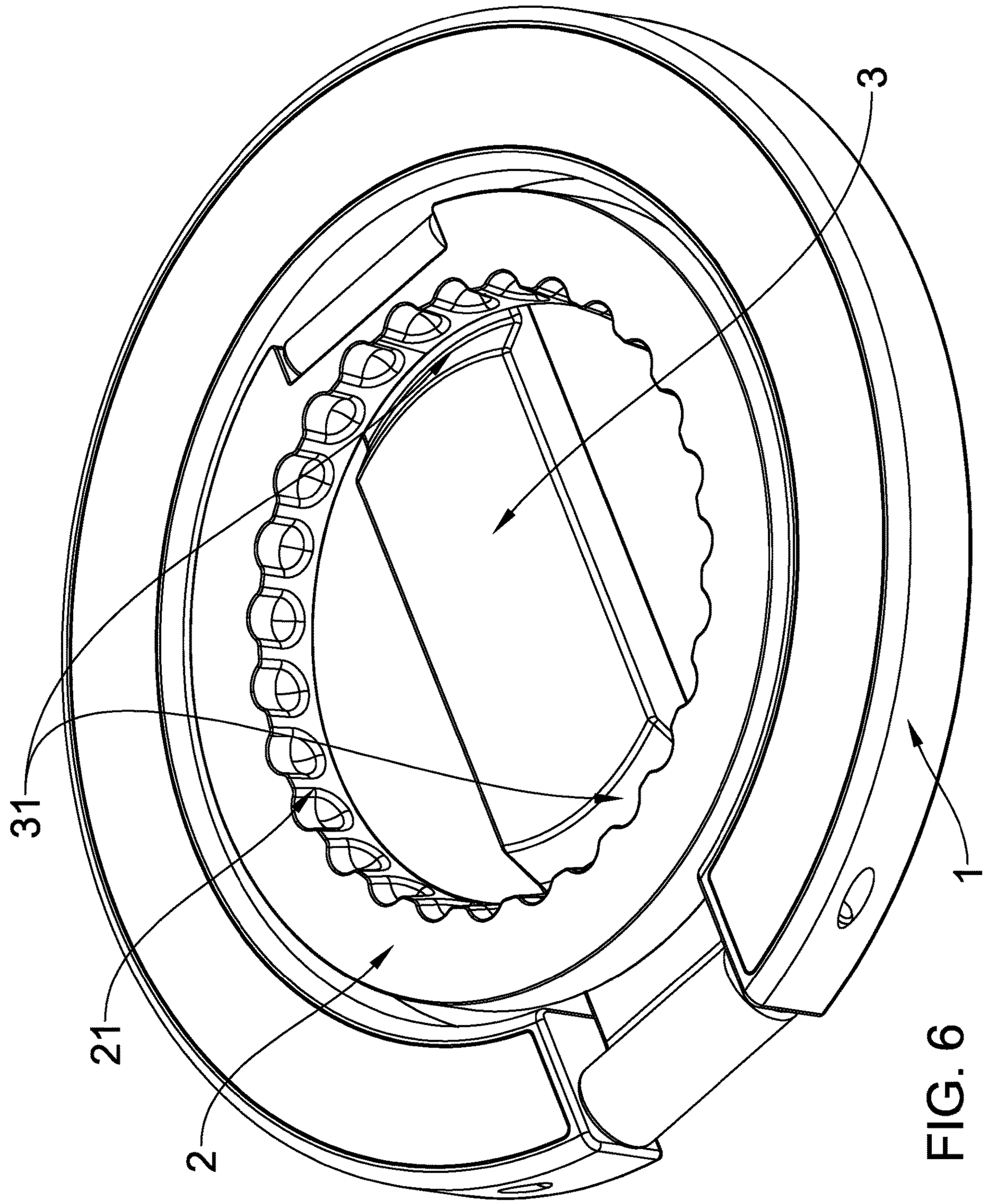


FIG. 6

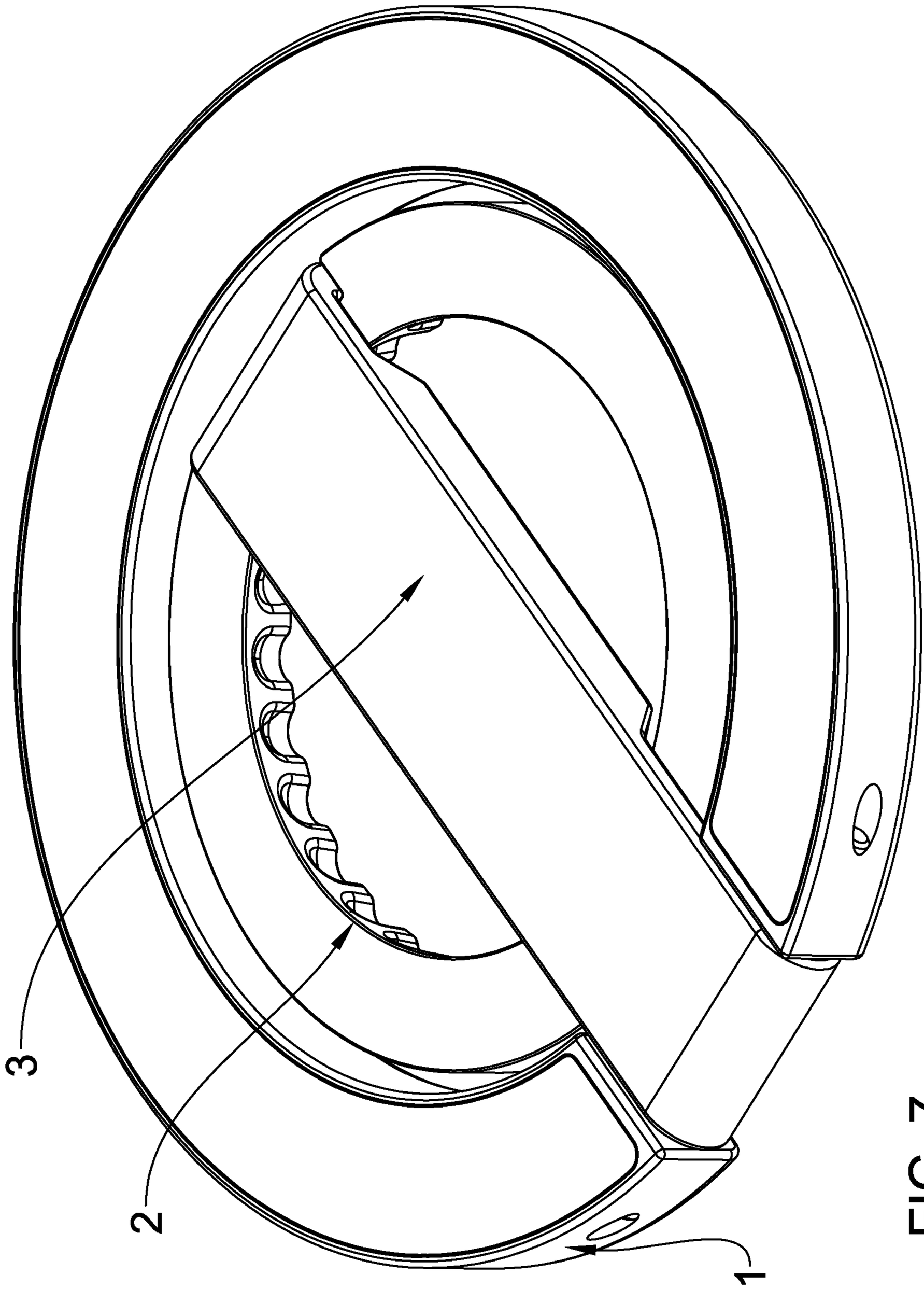


FIG. 7

TWO-IN-ONE MULTI-FUNCTIONAL MAGNETIC BRACKET

BACKGROUND OF THE INVENTION

1. Field of the Invention

The invention relates to magnetic brackets and more particularly to a two-in-one multi-functional magnetic bracket.

2. Description of Related Art

Conventionally, a magnetic bracket is attached to a mobile phone in a magnetic manner to support a mobile phone on a desktop. However, the conventional magnetic bracket has the following deficiencies during use:

Most conventional magnetic brackets on the market only have a function of supporting the mobile phone on the desktop, with a single function, and can be switched according to different scenarios and platforms. When being used on a center console in a car or in an office scenario, the magnetic bracket is suspended on a side of a desktop computer. However, in comparison of information, a special magnetic bracket needs to be bought to fix the mobile phone. Therefore, a user needs to prepare a plurality of magnetic brackets with different functions. This is inconvenient to carry and raises an economic expenditure, and brings a serious trouble to the user.

Thus, the need for improvement still exists.

SUMMARY OF THE INVENTION

To achieve the above objective, the invention provides the following technical solutions: A two-in-one multi-functional magnetic bracket comprises a first bracket, where a magnet is embedded in an inner part of the first bracket, to enable the first bracket to be magnetically attached to a mobile phone; a second bracket, where the second brackets movably connected to the first bracket through a connecting arm, the second bracket and a hinging arm are accommodated to the inner part of the first bracket by flipping, and a ring of tooth slot are opened in an inner part of the second bracket; and a connector, snapped in the inner part of the second bracket and combined with the second bracket to form a multi-platform bracket for mounting the second bracket on different scenarios and platforms, where the connector includes a base of which a surface is adhered to the platform, a limiting buckle that is disposed in an inner part of the base and of which one end extends to a side of the second bracket, and a spring bead that is disposed in the base and of which one end extends to an inner side of the tooth slot, when the limiting buckle is pulled away from the second bracket, the second bracket pulled, to separate the second bracket from the connector.

In a preferred technical solution of the invention, the first bracket is in an annular shape, and an accommodation hole is opened in the first bracket, to accommodate the second bracket and a wireless charger, a first notch is opened on a surface of the first bracket, and a first rotation axis is disposed in an inner part of the first notch.

In a preferred technical solution of the invention, the second bracket is in an annular shape, and a second notch is opened on a surface of the second bracket, a second rotation axis is disposed in an inner part of the second notch, one end of the hinging arm extends to the inner part of the first notch and is snapped to a surface of the first rotation axis, and the other end of the hinging arm extends to the inner part of the second notch and is snapped to a surface of the second rotation axis.

In a preferred technical solution of the invention, a placement slot in which the first brackets placed is opened on a side of the hinging arm, a slip resistant pad that is used to cover the magnet is disposed on one side of the first bracket, and a decorative sticker is disposed on the other side of the first bracket.

In a preferred technical solution of the invention, at least one first movable slot is disposed in the inner part of the base at equal intervals in an annular shape, the limiting buckle is movably disposed in an inner part of the first movable slot, a quantity of the limiting buckles is equal to a quantity of the first movable slots, and a pressure spring of which one end abuts against one phase of the limiting buckle is disposed in the inner part of the first movable slot.

In a preferred technical solution of the invention, a slant block is disposed at one end of a limiting buckle located in the movable slot, a snap-fastener is movably disposed on a side of the base, a press block corresponding to the slant block is disposed on the snap-fastener, when the snap-fastener is pressed, the snap-fastener drives the press block to abut against the slant block, and the slant block is abutted to drive the limiting buckle to compress the pressure spring and separate from the second bracket.

In a preferred technical solution of the invention, a cover plate is disposed on the side of the base, a through hole is opened in an inner part of the cover plate, and a side of the snap-fastener extends to an inner part of the through hole.

In a preferred technical solution of the invention, at least one first movable slot is disposed in the inner part of the base at equal intervals in an annular shape, the spring bead is movably disposed in an inner part of the first movable slot, and a quantity of the spring beads is equal to a quantity of the second movable slots.

In a preferred technical solution of the invention, a clamping groove of which one end extends to the inner part of the first movable slot and is clamped with the spring bead is disposed on a side of the cover plate.

In a preferred technical solution of the invention, an adhesive mat is disposed on a side, far away from the cover plate, of the base.

The invention has the following advantages and benefits in comparison with a single-function magnetic bracket of the conventional art:

The two-in-one multi-functional magnetic bracket can be used on a desktop, and can be suspended on a side of an office computer or mounted in a center console of a car through the base. When the connector is separated from and the second bracket, the second bracket can be sleeved in a finger as a finger ring, to support the product by hands. When the two-in-one multi-functional magnetic first brackets used on the desktop, the two-in-one multi-functional magnetic bracket can be adjusted and supported at 360°. The two-in-one multi-functional magnetic first brackets applicable to a variety of scenarios and platforms, to ensure usability of the product. The product has an application prospect.

The above and other objects, features and advantages of the invention will become apparent from the following detailed description taken with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of a two-in-one multi-functional magnetic bracket according to the invention;

FIG. **2** is a view similar to FIG. **1** with the connector separated;

FIG. **3** is a view similar to FIG. **2** with the connector removed and the first bracket shown in an exploded manner;

FIG. **4** is an exploded view of the connector;

FIG. **5** is another exploded view of the connector;

FIG. **6** is a perspective view of the folded two-in-one multi-functional magnetic bracket; and

FIG. **7** is another perspective view of the folded two-in-one multi-functional magnetic bracket.

DETAILED DESCRIPTION OF THE INVENTION

Referring to FIGS. **1** to **7**, a two-in-one multi-functional magnetic bracket of the invention comprises the following components as discussed in detail below.

A first bracket **1** is provided, where a magnet **14** is embedded in an inner part of the first bracket **1**, to enable the first bracket **1** to be magnetically attached to a mobile phone.

A second bracket **2** is provided, where the second bracket **2** is movably connected to the first bracket **1** through a connecting arm, the second bracket **2** and a hinging arm **3** are accommodated to the inner part of the first bracket **1** by flipping, and a ring of tooth slot **21** are opened in an inner part of the second bracket **2**.

A connector **4** is, snapped in the inner part of the second bracket **2** and combined with the second bracket **2** to form a multi-platform bracket for mounting the second bracket **2** on different scenarios and platforms, where the connector **4** includes a base **41** of which a surface is adhered to the platform, a limiting buckle **44** that is disposed in an inner part of the base **41** and of which one end extends to a side of the second bracket **2**, and a spring bead **431** that is disposed in the base **41** and of which one end extends to an inner side of the tooth slot **21**, when the limiting buckle **44** is pulled away from the second bracket **2**, the second bracket **2** is pulled, to separate the second bracket **2** from the connector **4**.

In a specific technical solution of the embodiment, the first bracket **1** is in an annular shape, and an accommodation hole **11** is opened in the first bracket **1**, to accommodate the second bracket **2** and a wireless charger. A first notch **12** is opened on a surface of the first bracket **1**, and a first rotation axis **33** is disposed in an inner part of the first notch **12**.

As shown in FIGS. **6** and **7** specifically, the first bracket **1** is attached to the mobile phone by using the magnet **14**, and the second bracket **2** is flipped out of the inner part of the accommodation hole **11** through the hinging arm **3**, so that the wireless charger may be placed in the inner part of the accommodation hole **11**. This is more convenient to use.

In a specific technical solution of the embodiment, the second bracket **2** is in an annular shape, and a second notch **22** is opened on a surface of the second bracket **2**, a second rotation axis **23** is disposed in an inner part of the second notch **22**, one end of the hinging arm **3** extends to the inner part of the first notch **12** and is snapped to a surface of the first rotation axis **33**, and the other end of the hinging arm **3** extends to the inner part of the second notch **22** and is snapped to a surface of the second rotation axis **23**. Through the first rotation axis **33** and the second rotation axis **23**, the first bracket **1**, the hinging arm **3**, the second bracket **2**, and the hinging arm **3** may rotate, to facilitate the adjustment of a support angle.

In a specific technical solution of the embodiment, a placement slot **31** in which the second bracket **2** is placed is opened on a side of the hinging arm **3**. A slip resistant pad **15** that is used to cover the magnet **14** is disposed on one side of the first bracket **1**. A decorative sticker **16** is disposed on the other side of the first bracket **1**. The slip resistant pad **15** is made of a rubber material. Therefore, when being attached to a mobile phone of a user, the slip resistant pad **15** is not worn, slip resistant effect is excellent, and adsorption is more stable. The decorative sticker **16** is used to decorate the side of the first bracket **1**, so that appearance is more beautiful.

In a specific technical solution of the embodiment, at least one first movable slot **42** is disposed in the inner part of the base **41** at equal intervals in an annular shape. The limiting buckle **44** is movably disposed in an inner part of the first movable slot **42**. A quantity of the limiting buckles **44** is equal to a quantity of the first movable slots **42**. A pressure spring **46** of which one end abuts against one phase of the limiting buckle **44** is disposed in the inner part of the first movable slot **42**. The first movable slot **42** is used to mount the limiting buckle **44**. Therefore, the limiting buckle **44** may be more stably retracted, and when the limiting buckle **44** slides toward the inner part of the first movable slot **42**, the limiting buckle **44** may compress the pressure spring **46**, and while the compression spring **46** may continuously abut against the limiting buckle **44**. Therefore, when the user releases a snap-fastener **47**, the compression spring **46** springs back and abuts against the limiting buckle **44**, to enable the limiting buckle **44** to automatically move toward an outer part of the first movable slot **42**.

In a specific technical solution of the embodiment, a slant block **45** is disposed at one end of a limiting buckle **44** located in the movable slot. The snap-fastener **47** is movably disposed on a side of the base **41**. A press block **471** corresponding to the slant block **45** is disposed on the snap-fastener **47**. A cover plate **48** is disposed on the side of the base **41**, a through hole **481** is opened in an inner part of the cover plate **48**, and a side of the snap-fastener **47** extends to an inner part of the through hole **481**. When the snap-fastener **47** is pressed, the snap-fastener **47** drives the press block **471** to abut against the slant block **45**, and the slant block **45** is abutted to drive the limiting buckle **44** to compress the pressure spring **46** and separate from the second bracket **2**.

As shown in FIGS. **4** and **5** specifically, the pressure block **471** abuts against a beveled surface on the slant block **45**. When the snap-fastener **47** is pressed, the snap-fastener **47** drives the pressure block **471** to vertically move down and abut against the slant block **45**. Therefore, when being abutted, the beveled surface on the slant block **45** compresses the pressure spring **46** and slides along pressure block **471**. When the snap-fastener **47** is released, the pressure spring **46** springs back to drive the slant block **45** to be against the pressure block **471**, to enable the pressure block **471** to drive the snap-fastener **47** to enter the inner part of the through hole **481**. This facilitates pressing by the user. Because a beveled surface is disposed on the limiting buckle **44**, when the second bracket **2** is assembled to the connector **4**, when the beveled surface of the limiting buckle **44** is directly abutted, the limiting buckle **44** is driven to be retracted to the inner part of the first movable slots **42**. The snap-fastener **47** does not need to be pressed, and only needs to be pressed during disassembly.

In a specific technical solution of the embodiment, at least one second movable slot **43** is disposed in the inner part of the base **41** at equal intervals in an annular shape, the spring bead **431** is movably disposed in an inner part of the second movable slot **43**, and a quantity of the spring beads **431** is equal to a quantity of the second movable slots **43**. The second movable slot **43** is used to mount the spring bead **431**. The spring bead **431** includes a bead and a straight barrel that sleeves the bead. A spring is disposed in an inner

part of the straight barrel. When the second bracket **2** drives the tooth slot **21** to rotate on the base **41**, the tooth slot **21** is frequently switched to abut against the spring bead **431**. When the bead of the spring bead **431** is abutted, the spring is compressed backward, and clamped into an inner part of a next tooth slot **21** while the tooth slot **21** is switched, to position the rotated second bracket **2**.

In a specific technical solution of the embodiment, a clamping groove **482** of which one end extends to the inner part of the second movable slot **43** and is clamped with the spring bead **431** is disposed on a side of the cover plate **48**. Due to the clamping groove **482**, stabilization of mounting the spring bead **431**.

In a specific technical solution of the embodiment, an adhesive mat **49** is disposed on a side, far away from the cover plate **48**, of the base **41**. The base **41** may be adhered to a different scenario such as an object or a platform through the adhesive mat **49**.

During use in a car, the base **41** is adhered to a center console of the car through the adhesive mat **49**, and the first bracket **1** is attached to the mobile phone through the magnet **14**. The hinging arm **3** and the second bracket **2** are flipped out from the inner part of the accommodation hole **11**, and the second bracket **2** is flipped out from an inner part of the placement slot **31**, and the second bracket **2** is sleeved on the connector **4**. The second bracket **2** abuts against the limiting buckle **44**. When the limiting buckle **44** is abutted, the spring is compressed backward. When the second bracket **2** crosses the limiting buckle **44**, the limiting buckle **44** is abutted by rebound of the spring and reset to clamp the second bracket **2**, to steadily sleeve the second bracket **2** on the connector **4**. When an angle of the second bracket **2** needs to be adjusted, the second bracket **2** can be rotated at 360° on the surface of the base **41**. After being rotated, the second bracket **2** can be clamped into the tooth slot **21** to position. In the office scenario, the base **41** can be adhered to a back of a screen of a desktop computer or a laptop through the adhesive mat **49**. Therefore, the mobile phone is suspended on a side of the computer, so that the user can conveniently view information. Alternatively, the base **41** can be mounted on the desktop for use, or the connector **4** can be separated from the second bracket **2**. Therefore, the second bracket **2** can be sleeved in a finger of the user, to realize handheld support. Alternatively, the second bracket **2** can be accommodated in the accommodation hole **11**.

The invention has the following advantages and benefits in comparison with a single-function magnetic bracket of the conventional art: the two-in-one multi-functional magnetic bracket can be used on a desktop, and can be suspended on a side of an office computer or mounted in a center console of a car through the base. When the connector is separated from and the second bracket, the second bracket can be sleeved in a finger as a finger ring, to realize handheld support. When the two-in-one multi-functional magnetic first brackets used on the desktop, the two-in-one multi-functional magnetic bracket can be adjusted and supported at 360°. The two-in-one multi-functional magnetic first brackets applicable to a variety of scenarios and platforms, to ensure usability of the product. The product has an application prospect.

While the invention has been described in terms of preferred embodiments, those skilled in the art will recognize that the invention can be practiced with modifications within the spirit and scope of the appended claims.

What is claimed is:

1. A multi-functional magnetic bracket, comprising:

a first bracket, wherein a magnet is embedded in an inner part of the first bracket, to enable the first bracket to be magnetically attached to a mobile phone;

a second bracket, wherein the second bracket is movably connected to the first bracket through a connecting arm, the second bracket and a hinging arm are accommodated to the inner part of the first bracket by flipping, and a ring and a tooth slot are opened in an inner part of the second bracket; and

a connector, snapped in the inner part of the second bracket and combined with the second bracket to form a multi-platform bracket for mounting the second bracket on different scenarios and platforms, wherein the connector comprises a base of which a surface is adhered to the platform, a limiting buckle that is disposed in an inner part of the base and of which one end extends to a side of the second bracket, and a spring bead that is disposed in the base and of which one end extends to an inner side of the tooth slot, when the limiting buckle is pulled away from the second bracket, the second bracket is pulled, to separate the second bracket from the connector.

2. The multi-functional magnetic bracket of claim **1**, wherein the first bracket is in an annular shape, and an accommodation hole is opened in the first bracket, to accommodate the second bracket and a wireless charger, a first notch is opened on a surface of the first bracket, and a first rotation axis is disposed in an inner part of the first notch.

3. The multi-functional magnetic bracket of claim **2**, wherein the second bracket is in an annular shape, and a second notch is opened on a surface of the second bracket, a second rotation axis is disposed in an inner part of the second notch, one end of the hinging arm extends to the inner part of the first notch and is snapped to a surface of the first rotation axis, and the other end of the hinging arm extends to the inner part of the second notch and is snapped to a surface of the second rotation axis.

4. The multi-functional magnetic bracket of claim **3**, wherein a placement slot in which the second bracket is placed is opened on a side of the hinging arm, a slip resistant pad that is used to cover the magnet is disposed on one side of the first bracket, and a decorative sticker is disposed on the other side of the first bracket.

5. The multi-functional magnetic bracket of claim **1**, wherein at least one first movable slot is disposed in the inner part of the base at equal intervals in an annular shape, the limiting buckle is movably disposed in an inner part of the first movable slot, a quantity of the limiting buckles is equal to a quantity of the first movable slots, and a pressure spring of which one end abuts against one phase of the limiting buckle is disposed in the inner part of the first movable slot.

6. The multi-functional magnetic bracket of claim **1**, wherein a slant block **45** is disposed at one end of a limiting buckle located in the first movable slot, a snap-fastener is movably disposed on a side of the base, a press block corresponding to the slant block is disposed on the snap-fastener, when the snap-fastener is pressed, the snap-fastener drives the press block to abut against the slant block, and the slant block is abutted to drive the limiting buckle to compress the pressure spring and separate from the second bracket.

7. The multi-functional magnetic bracket of claim **6**, wherein a cover plate is disposed on the side of the base, a through hole is opened in an inner part of the cover plate, and a side of the snap-fastener extends to an inner part of the through hole.

8. The multi-functional magnetic bracket of claim 7, wherein at least one second movable slot is disposed in the inner part of the base at equal intervals in an annular shape, the spring bead is movably disposed in an inner part of the second movable slot, and a quantity of the spring beads is equal to a quantity of the second movable slots.

9. The multi-functional magnetic bracket of claim 8, wherein a clamping groove of which one end extends to the inner part of the second movable slot and is clamped with the spring bead is disposed on a side of the cover plate.

10. The multi-functional magnetic bracket of claim 9, wherein an adhesive mat is disposed on a side, far away from the cover plate of the base.

* * * * *